When the Grass Is Greener

Three-Shot Search on an Exponentiated Ornstein–Uhlenbeck Landscape

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First version: April 6, 2022. This version: September 3, 2026.

Abstract

How should a firm spend its last pivot? A searcher knows an incumbent strategy's performance, may run one more trial, and must then commit, on a landscape given by the exponential of a stationary Ornstein–Uhlenbeck path. We solve the final placement exactly and give a closed-form phase diagram for the case of equal observations: abandon a below-median incumbent, leave past the better position after a disappointing trial, commit between two good positions, and stay put above an exact ceiling. The commitment between positions rests on a constant-variance identity, that an interior position has the same expected performance as a bolder alternative and $(1 + \rho)/(1 - \rho)$ times its variance. The same rule, used as the line search inside a derivative-free optimizer, beats classical inner loops on rough and high-dimensional objectives in a public benchmark.

1 Introduction

How many pivots does a firm get? Across four randomized trials covering 759 startups, 59% never pivoted radically during the observation window, about 30% pivoted exactly once, and 11% pivoted more than once (Camuffo et al. 2024; the shares in the smaller 2020 trial are 74%, 19%, and 7%). These are one-year observation windows; at the implied rate of roughly half a pivot per year, a startup's three-to-five-year life carries a budget on the order of two or three pivots in total. Training founders to search more scientifically raised the frequency of a single pivot and lowered the frequency of repeated pivots. In practice, strategic search involves very few evaluations.

Callander (2011) modeled this kind of search as trial and error on an unknown function drawn from a Brownian motion. A sequence of short-lived agents each implements one trial and observes its outcome, so search is myopic by demographic structure rather than by assumption. Most later work keeps this feature. Bardhi and Callander (2026) list farsighted search with a free choice of location as an open problem.

We solve the smallest such problem exactly. The landscape is the exponential of a stationary Ornstein–Uhlenbeck path. Mean reversion means that performance far from known territory regresses to the industry average. The exponential means that small differences in fit produce large differences in sales, so an untried strategy has extra value on average.

The searcher knows the performance of an incumbent strategy. She may try one more. Then she must commit. With two evaluations the optimal rule is a threshold rule: abandon a below-median incumbent, keep an exceptional one, and otherwise step a distance that we give in closed form.

With three evaluations the question is where to commit after the trial: between the two known strategies, beyond the better one, or at one of them. We compute the answer exactly. Commitment between the two is optimal when both values are good but not exceptional. The commitment point is the midpoint for moderate values and moves toward the better-known end as the values improve. Above an exact ceiling the searcher should stay where she is.

The interior option is worth about one percent of expected value at the optimum in our calibration. Most of the value of the final pivot comes from the information it produces. On landscapes that decorrelate quickly, it is better to probe far away, and the formulas say so: the interior premium is first order in the bracket correlation.

The rule is also an algorithm, and the direction of trade is unusual: a policy derived in the economics of search outperforms classical line searches on their own ground. Used as the inner loop of a derivative-free optimizer, the three-evaluation rule beats golden-section and Brent on collision and agent-chaos objectives and on high-dimensional ones, where its one- or two-evaluation lines buy several times more search directions per budget (Section 10).

2 Related work

Callander (2008) introduced the Brownian policy landscape. In that paper, delegation to a biased expert survives only when the bias is below $\sigma^2/2\mu$. The 2011 papers use the landscape for search. An agent experiments when the incumbent outcome is worse than $\alpha = \sigma^2/2|\mu|$ and stops otherwise. Search ends almost surely, often at a mediocre policy. Simulated runs change policy between one and four times, which is consistent with the pivot counts above.

Later papers derive similar thresholds for precedent (Callander and Clark 2017), agenda control (Callander and McCarty 2024), and risk-averse learning (Callander and Matouschek 2019). In all of these the number of evaluations is endogenous and payoffs accrue every period. In our problem the number of evaluations is fixed at three, only the final placement is paid, and the landscape mean-reverts.

Callander and Hummel (2014) analyze a two-period game in which each of two policymakers can experiment once. The equilibrium has two exact cutoffs, and the first mover sometimes experiments although the status quo already delivers his ideal outcome. The players are strategic and payoffs accrue in both periods. Ganz (2020) studies an organization that allows a manager one offline trial before an online choice. That is a two-evaluation problem in which only the second evaluation is paid, and some of its boundaries are computed numerically. We solve the three-evaluation single-agent problem: the final placement exactly and, for equal observed values, with every phase boundary in closed form; the opening trial reduces to a one-dimensional numerical optimization.

Other papers obtain farsighted results under other restrictions. Urgun and Yariv (2025) require search to be contiguous; the searcher chooses speed and stops at a closed-form drawdown. Cetemen, Urgun and Yariv (2023) extend this to teams and Wong (2025) to flow payoffs. Bardhi (2024) chooses k sample locations at once to estimate a project. Garfagnini and Strulovici (2016), Carnehl and Schneider (2025), and Callander, Lambert and Matouschek (forthcoming) study infinite sequences of short-lived agents.

Aybas and Callander (2025) allow Ito diffusion landscapes, including the Ornstein–Uhlenbeck case, in a one-shot cheap-talk game. That is the only mean-reverting landscape we know of in this literature. Bardhi (2024) also proves that the OU kernel is the only powered-exponential covariance with the Markov property, which supports our choice of landscape.

In the optimization literature the same object appears with large budgets. Calvin (1997) derives average-case convergence rates under Wiener measure. Grill, Valko and Munos (2018) approximate the maximum of a Brownian path. Grosse, Zhang and Hennig (2023) extend this to Gaussian-process samples and prove regret rates. Alabert and Caballero (2018) compute the law of the minimum of a conditioned bridge in order to compare fixed sampling designs.

Malladi (2025) and Banchio and Malladi (2025) solve worst-case versions with no prior and a Lipschitz bound in place of volatility. Hodgson and Lewis (2025) estimate a Gaussian-process search model on clickstream data and solve the policy numerically. The probabilistic line search of Mahsereci and Hennig (2017) is the nearest algorithmic relative: a Gaussian-process line search for stochastic gradient descent, built on gradient observations and an integrated-Wiener model, without closed-form placement. We are not aware of a closed-form solution for a small fixed budget of adaptive, derivative-free evaluations under a Bayesian prior.

The closest methodological neighbor is non-myopic Bayesian optimization. Ginsbourger and Le Riche (2010) observed that the usual one-step expected-improvement rule is suboptimal under a fixed finite budget, since it treats each evaluation as the last. Lam, Willcox and Wolpert (2016) cast finite-budget optimization as a dynamic program and solve it approximately by rollout, and Jiang et al. (2020) optimize the full multi-step lookahead tree in one shot. These methods approximate the lookahead dynamic program on general Gaussian-process priors; we solve the smallest nontrivial case, three evaluations on an OU landscape, exactly. The knowledge-gradient policy of Frazier, Powell and Dayanik (2009) is the closest on payoff, since it scores the improvement in the finally recommended point, a terminal-recommendation objective like ours, though it looks one step ahead.

The rule is itself an acquisition function, and it sits at a specific place in the standard taxonomy (Shahriari et al. 2016). Acquisition functions sort on two axes. One is myopic versus lookahead: expected improvement (Mockus et al. 1978; Jones et al. 1998), probability of improvement (Kushner 1964), the upper confidence bound (Srinivas et al. 2010), and the entropy-search family (Hennig and Schuler 2012; Wang and Jegelka 2017) are one-step rules, while multi-step-optimal acquisitions exist only in approximate form. The other is cumulative-regret versus terminal-recommendation payoff: the upper confidence bound bounds cumulative regret, whereas knowledge gradient and entropy search score a finally reported point. Our rule is lookahead and terminal-recommendation, the knowledge-gradient corner, and it fills the cell that is otherwise empty, an acquisition that is exact and closed-form for a small multi-step budget rather than one-step or approximate. A recurring finding in this literature is that multi-step lookahead buys only modest gains over the myopic rule (Wu and Frazier 2019); the one-percent value of the interior option in Section 9 is that gain computed exactly rather than measured.

Glick and Myers (2015) tested the one-evaluation problem on lab subjects. Subjects respond to the Brownian structure but modify too much when the variance is high.

3 The model: three evaluations of an exponentiated OU path

Let X be a stationary Ornstein–Uhlenbeck process on $\mathbb{R}$ with $\mathbb{E}[X_t] = 0$, unit marginal variance, and covariance kernel

$$k(s,t) \;=\; e^{-\kappa|s-t|}. \tag{1}$$

Exponential decay of correlation with distance is a standard empirical model, not an assumption special to this paper: it has described shadow fading in radio engineering since Gudmundson (1991). On 77,040 WiFi measurements across a university floor (Feng, Nguyen and Luo 2024), we find the detrended log signal close to Gaussian with spatial correlation fitting an exponential at $R^2 = 0.98$, alongside a short-scale component of comparable size; the scripts are in the paper's repository.

The same shape is standard in several other fields. In Gaussian-process modeling the exponential kernel is the Matérn class at $\nu = 1/2$ and is the covariance of the OU process (Rasmussen and Williams 2006, §4.2.1). Practical Bayesian optimization prefers Matérn kernels because squared-exponential sample paths are too smooth for real objectives (Snoek, Larochelle and Adams 2012).

Weinberger (1990) measures the ruggedness of a fitness landscape by the exponential decay rate of a random walk's autocorrelation and shows that Kauffman's NK model is generically of this type. In evolutionary biology the OU process is the standard model of a trait held near an adaptive optimum (Hansen 1997; Butler and King 2004). The landscape is $f(t) = \exp(X_t)$: locally continuous, mean-reverting in its logarithm, with occasional high peaks where excursions are magnified by the exponential. A searcher samples f at a first location, which we take to be $t_0 = 0$ by translation invariance, discovering $X_0 = b$. Two further evaluations at locations t_1 and then t_2 follow, each chosen with knowledge of all values so far, and the payoff is the value at the final location, $u = \mathbb{E}[f(t_2)]$. Conditional on the observations gathered by the time t_2 is chosen, X_{t_2} is Gaussian with some mean μ and variance ν, and

$$\log \mathbb{E}\left[e^{X_{t_2}} \,\middle|\, \mathcal{F}\right] \;=\; \mu + \tfrac{1}{2}\nu, \tag{2}$$

where $\mathcal{F}$ denotes the observed locations and values. (The unconditional distribution after adaptive placement need not be Gaussian; every use of (2) below is conditional.) The searcher is thus paid for conditional mean and conditional variance in equal measure: a final location with residual uncertainty carries a lognormal bonus of half its variance.

Figure 1 shows the decision this paper solves.

The problem has one essential scale parameter. A payoff exponent β alone is without loss of generality, since $e^{\beta X}$ on a unit-variance path equals e^{Y} on a path of standard deviation β; only the product $s = \beta\sigma$ of payoff convexity and path scale matters. We calibrate $s = 1$.

For general s every comparison maximizes $\mu + s\nu/2$ with observations in path-standard-deviation units, the two-shot step becomes $\lambda^* = \mathrm{clip}(b/s, 0, 1)$, and the pitchfork and ceiling of Section 7 scale to $s \cdot 2\sqrt{\rho}/(1-\rho)$ and $s(1+\rho)/(1-\rho)$; equivalently, the phase diagram is universal in the coordinates $(\rho, b/s)$. The length scale κ, by contrast, is fully absorbed by working in correlations. The parameter plays the role Callander's complexity $\sigma^2/2|\mu|$ plays under drift, and the one-percent figure of Section 9 is specific to $s = 1$.

Two conventions tidy the boundary cases. A searcher may take a location to infinity, which by mean reversion is equivalent to sampling a fresh $N(0,1)$ value independent of everything seen; and a searcher may finish at an observed location, receiving its known value. Neither changes the substance of any result.

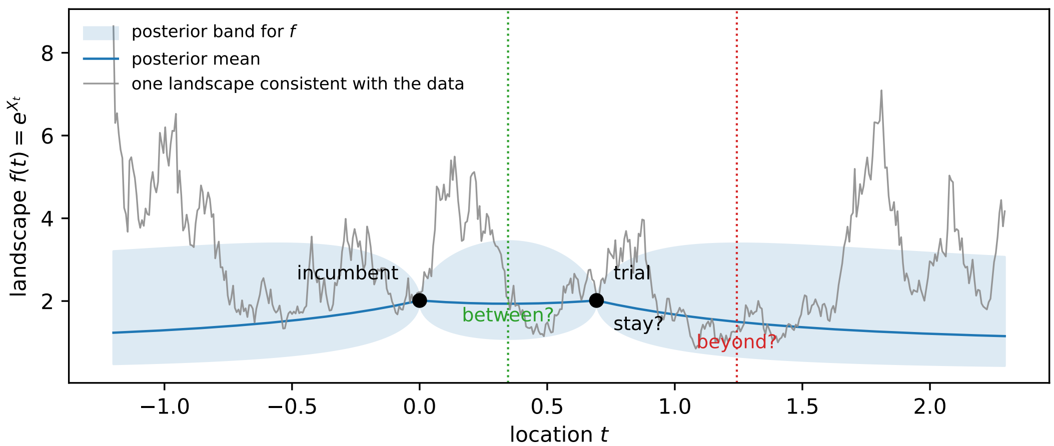

Figure 1: The problem. Two locations have been evaluated and returned equal values (dots); the shaded region is the posterior band for the landscape and the grey curve one landscape consistent with the data. The final evaluation is the consequential one: settle between the anchors, step out past an end, or stay at an observed point. The paper computes which, exactly, as a function of the anchor value and the bracket correlation.

4 Two evaluations: flee, explore, or stay

Given only $X_0 = b$, a candidate second location at distance t has $X_t \sim N\big(b\lambda,\, 1-\lambda^2\big)$ with $\lambda = e^{-\kappa t} \in (0, 1]$. From (2),

$$\log \mathbb{E}\big[e^{X_t} \,\big|\, X_0 = b\big] \;=\; b\lambda + \tfrac{1}{2}\big(1-\lambda^2\big) \;=\; -\tfrac{1}{2}\,(\lambda - b)^2 + \tfrac{1}{2}\,\big(b^2+1\big)\,, \tag{3}$$

maximized over $\lambda \in [0, 1]$ at $\lambda^* = \min\{\max\{b, 0\}, 1\}$.

Proposition 1 (Two-shot policy). The optimal second location and its log-value $\zeta(b)$ are

$$t^* \;=\; \begin{cases} \infty & b < 0 \quad \text{(flee)}, \\ -\frac{1}{\kappa}\log b & 0 \le b \le 1 \quad \text{(explore)}, \\ 0 & b > 1 \quad \text{(stay)}, \end{cases} \qquad \zeta(b) \;=\; \begin{cases} \frac{1}{2} & b < 0, \\ \frac{1}{2}\,\big(b^2+1\big) & 0 \le b \le 1, \\ b & b > 1. \end{cases} \tag{4}$$

A below-median first value is abandoned, and an exceptional one is kept: already at two shots, an observed location can be the optimal final location. Practitioners know this fork as pivot or persevere (Ries 2011); the proposition supplies its thresholds and the size of the step when pivoting wins. In between, the searcher steps to the distance at which the prior pull toward the mean exactly balances the anchor's height. The value ζ serves as the exterior benchmark throughout: by the Markov property, any placement beyond all existing observations faces a fresh copy of this problem anchored at the nearest observation.

5 Bridge moments, and what composes how

Suppose values $X_0 = b$ and $X_{t_1} = d$ are known, with bracket correlation $\rho = e^{-\kappa t_1}$ and rapidity $\Theta = \operatorname{artanh}\rho$. For an interior location let $\lambda_2 = e^{-\kappa t_2}$ and $\lambda = e^{-\kappa(t_1-t_2)}$ be the correlations to the two ends, $\lambda_2\lambda = \rho$, with rapidities $\theta_2 = \operatorname{artanh}\lambda_2$ and $\theta = \operatorname{artanh}\lambda$. Gaussian conditioning gives the bridge moments

$$\mu \;=\; \frac{\lambda_2\big(1-\lambda^2\big)\,b + \lambda\big(1-\lambda_2^2\big)\,d}{1-\rho^2}, \qquad \nu \;=\; \frac{\big(1-\lambda_2^2\big)\,\big(1-\lambda^2\big)}{1-\rho^2}. \tag{5}$$

For $b = d > 0$ the bridge sags: the mean is below b everywhere strictly inside, pulled toward the process mean, while the variance rises from zero at the ends to a maximum at the middle.

A caution on what is additive. Correlations across consecutive intervals multiply, $\rho(t+s) = \rho(t)\,\rho(s)$, so the additive coordinate for composing intervals is $\kappa t = -\log\rho$, not the rapidity. The rapidity enters through a different identity: for equal anchors $d = b$ the bridge mean is

$$\mu \;=\; b\,\frac{\lambda_2+\lambda}{1+\lambda_2\lambda} \;=\; b\tanh(\theta_2+\theta)\,, \tag{6}$$

which matches the mean of the exterior point at correlation $\tanh(\theta_2+\theta)$ to its anchor. This is a mean-matching device we use next, not a composition law. The bridge variance also takes a hyperbolic product form: substituting $1-\tanh^2 = \operatorname{sech}^2$ into (5),

$$\nu \;=\; \operatorname{sech}(\theta_2+\theta)\operatorname{sech}(\theta_2-\theta)\,, \tag{7}$$

while an exterior point at correlation $\lambda' = \tanh\theta'$ to its anchor has variance $1-\tanh^2\theta' = \operatorname{sech}^2\theta'$.

6 Matched-mean dominance at a constant variance ratio

Theorem 1 (Bridge dominance, equal anchors). Let $d = b$ and let $t_2 \in (0, t_1)$ be an interior location with rapidities (θ_2, θ). The exterior location at composed rapidity $\theta' = \theta_2 + \theta$ has the same conditional mean $b\tanh(\theta_2+\theta)$, and the ratio of conditional variances is the same at every interior point:

$$\frac{\nu_{\text{inside}}}{\nu_{\text{outside}}} \;=\; \frac{\cosh(\theta_2+\theta)}{\cosh(\theta_2-\theta)} \;=\; \frac{1+\tanh\theta_2\tanh\theta}{1-\tanh\theta_2\tanh\theta} \;=\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta} \;>\; 1. \tag{8}$$

Consequently the interior point's conditional distribution dominates its mean-matched exterior rival in the Gaussian convex order, and its expected payoff is at least as large for every convex payoff with finite expectation, strictly so for the exponential.

Proof. The mean statement is (6). The ratio of (7) to $\operatorname{sech}^2(\theta_2+\theta)$ is $\cosh(\theta_2+\theta)/\cosh(\theta_2-\theta)$, and expanding both via $\cosh(A\pm B) = \cosh A\cosh B \pm \sinh A \sinh B$ gives $(1+\tanh\theta_2\tanh\theta)/(1-\tanh\theta_2\tanh\theta)$, which is $(1+\rho)/(1-\rho)$ because $\tanh\theta_2\tanh\theta = \lambda_2\lambda = \rho$ on the bracket. Two Gaussians with equal means and ordered variances are ordered in the convex order (the wider equals the narrower plus independent noise), and (2) is the exponential case. □

The amplification is a property of the bracket, not the position: threefold at $\rho = \frac{1}{2}$, nineteenfold at $\rho = 0.9$, unbounded as $\rho \uparrow 1$. At the bracket's ends both variances vanish; their ratio still tends to $e^{2\Theta}$. Dominance over mean-matched exterior rivals, which the theorem explicitly constructs, decides nothing by itself: the exterior player may choose any correlation, and an observed anchor may simply be kept. The object that can fail to exist is different: an *unvisited* point whose conditional mean matches the best *observed* value. Mean reversion caps unvisited conditional means strictly below a high anchor, and whether the variance bonus makes up the shortfall is precisely the question the phase diagram answers. The next section makes this exact.

7 The constrained parabola and the complete symmetric solution

Parametrize the interior by $x = \lambda_2 \in (\rho, 1)$ and set

$$C \;=\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta}, \qquad m \;=\; \frac{x+\rho/x}{1+\rho} \;\in\; \left[\frac{2\sqrt{\rho}}{1+\rho},\, 1\right), \tag{9}$$

under which mirror points $x \leftrightarrow \rho/x$ coincide, the lower endpoint of m is the bracket middle, and, using $\left(1-x^2\right)\left(1-\rho^2/x^2\right) = (1+\rho)^2\left(1-m^2\right)$, the equal-anchor interior log-utility becomes a constrained parabola:

$$L_{\text{inside}}(m) \;=\; b\,m + \frac{C}{2}\left(1-m^2\right), \qquad m^* \;=\; \operatorname{clip}\!\left(\frac{b}{C};\; \frac{2\sqrt{\rho}}{1+\rho},\, 1\right). \tag{10}$$

This is the organizing object of the paper: the entire symmetric analysis reads off the position of the unconstrained peak b/C relative to the interval.

Theorem 2 (Pitchfork). The bracket middle is the interior optimum iff $b/C \le 2\sqrt{\rho}/(1+\rho)$, i.e.

$$b \;\le\; \frac{2\sqrt{\rho}}{1-\rho} \;=\; \sinh 2\tau, \qquad \tau = \operatorname{artanh}\sqrt{\rho}. \tag{11}$$

For $\sinh 2\tau < b < e^{2\Theta}$ the peak is interior and attained by the mirror pair

$$x_\pm \;=\; \frac{b\,(1-\rho) \pm \sqrt{b^2\,(1-\rho)^2 - 4\rho}}{2}, \qquad x_+x_- = \rho, \tag{12}$$

with optimal value

$$U_{\text{off}} \;=\; \frac{C}{2} + \frac{b^2}{2C} \;=\; \tfrac{1}{2}\left(b^2 e^{-2\Theta} + e^{2\Theta}\right). \tag{13}$$

For $b \ge e^{2\Theta}$ the peak $b/C \ge 1$ lies at or beyond the interval's open end: L_{inside} is increasing in m, no interior maximum is attained, and the interior supremum is the end limit b.

Proof. $dL/dm = b - Cm$ vanishes at $m = b/C$; the interval endpoints translate via (9) into the stated b-thresholds ($b/C = 2\sqrt{\rho}/(1+\rho)$ is $b = 2\sqrt{\rho}/(1-\rho)$, and $b/C = 1$ is $b = (1+\rho)/(1-\rho)$; the hyperbolic forms follow from $\tanh\tau = \sqrt{\rho}$). Solving $x + \rho/x = (1+\rho)\,b/C = b\,(1-\rho)$ gives (12), and substituting $m = b/C$ into (10) gives (13). □

Theorem 3 (Symmetric phase diagram). For $d = b \ge 0$ the best interior point strictly beats every alternative — exterior placements and observed anchors alike — precisely for

$$b_-(\rho) \;<\; b \;<\; e^{2\Theta}, \qquad b_-(\rho) = \frac{\sqrt{\rho}\left(2 - \sqrt{2\,(1-\rho)}\right)}{1+\rho}, \tag{14}$$

the lower edge being the smaller root of $b^2\,(1+\rho) - 4b\sqrt{\rho} + 2\rho = 0$. For $b \ge e^{2\Theta}$ the optimal final location is an *observed* one: staying at an anchor beats every unvisited point, interior or exterior.

Proof. If $b \geq e^{2\Theta}$, Theorem 2 gives interior supremum b, unattained, while every exterior point loses to the anchor pointwise: for $b > 1$ and any $\lambda \in [0, 1)$,

$$b - \left[b\lambda + \tfrac{1}{2}\left(1 - \lambda^2\right)\right] \;=\; (1 - \lambda)\left[b - \tfrac{1+\lambda}{2}\right] \;>\; 0 \tag{15}$$

(the suprema coincide at b; the dominance is strict at every unvisited point): staying at an anchor wins. If $\sinh 2\tau < b < e^{2\Theta}$ the interior optimum is U_{off}, and both comparisons are one line: for $b > 1$,

$$U_{\mathrm{off}} - b \;=\; \tfrac{1}{2}\, e^{-2\Theta}\left(b - e^{2\Theta}\right)^2 \;>\; 0, \tag{16}$$

the arithmetic–geometric mean inequality applied to the two terms of (13), with equality only at the tangency $b = e^{2\Theta}$ defining the upper edge; for $b \leq 1$,

$$U_{\mathrm{off}} - \tfrac{1}{2}\left(b^2 + 1\right) \;=\; \tfrac{1}{2}\left(e^{2\Theta} - 1\right)\left(1 - b^2 e^{-2\Theta}\right) \;>\; 0. \tag{17}$$

If $b \leq \sinh 2\tau$ the optimum is the middle, with value $U_{\mathrm{mid}} = b\,\frac{2\sqrt{\rho}}{1+\rho} + \frac{1-\rho}{2(1+\rho)}$; for $b \leq 1$, $U_{\mathrm{mid}} > \zeta(b)$ is the stated quadratic, and $b_- \leq 2\sqrt{\rho}/\left(1 + \rho\right) < \sinh 2\tau$ places the lower edge inside this regime. No gap opens above the middle regime: when $\sinh 2\tau \leq 1$ (i.e. $\rho \leq 3 - 2\sqrt{2}$), the larger quadratic root satisfies $b_+ \geq \sinh 2\tau$, equivalent to $(1 - \rho)^3 \geq 8\rho^2$, which holds on that range (left side decreasing, right increasing, valid at the endpoint); when $\sinh 2\tau > 1$, $b_+ \geq 1$ reduces, with $u = \sqrt{\rho}$, to $2u^2\left(1 + u\right) \geq (1 - u)^3$, true at $u = \sqrt{2} - 1$ (it reads $20\sqrt{2} \geq 28$) and increasingly so; and on $1 < b \leq \sinh 2\tau$, $U_{\mathrm{mid}} \geq b$ reduces to $b \leq (1 + u) / (2\,(1 - u))$, which covers the stretch because $\sinh 2\tau \leq (1 + u) / (2\,(1 - u))$ is $(1 - u)^2 \geq 0$. $\square$

Along the two-shot-inherited bracket $\rho = b$ the region opens at $b^* = 0.4233\ldots$, the root of $U_{\mathrm{mid}}(b, b) = \zeta(b)$. The geometry of the upper edge is easy to misread: for $b > 1$, interior dominance requires $\rho > (b - 1) / (b + 1)$, so higher anchors demand a larger bracket correlation, which is a *shorter* physical bracket. Very good news is forgiven only by narrow brackets; on a wide bracket, very good news means stay.

8 Beyond equal anchors

For general (b, d) the interior stationary points solve the quartic

$$x^4 - (b - \rho d)\, x^3 + \rho\,(d - \rho b)\, x - \rho^2 \;=\; 0, \tag{18}$$

obtained by differentiating (5); its real roots in $(\rho, 1)$, compared against $\zeta(b)$ and $\zeta(d)$, decide the move exactly and replace any grid. The symmetric window does not extend indefinitely in d. At $b = \frac{1}{2}$, $\rho = \frac{1}{2}$, the interior wins precisely for $d \in (\approx 0.4715, 1)$: below, the ray past the better end wins; at $d = 1$ every strictly interior point loses, by the identity

$$L(x) - 1 \;=\; -\frac{(2x - 1)^2\left(x^2 + x + 1\right)}{6x^2} \;<\; 0, \qquad \tfrac{1}{2} < x < 1, \tag{19}$$

and for $d \geq 1$ staying at the second anchor is optimal, since d's coefficient in the bridge mean is below one and raising d only widens the loss. Symmetric dominance is a window, not a half-line.

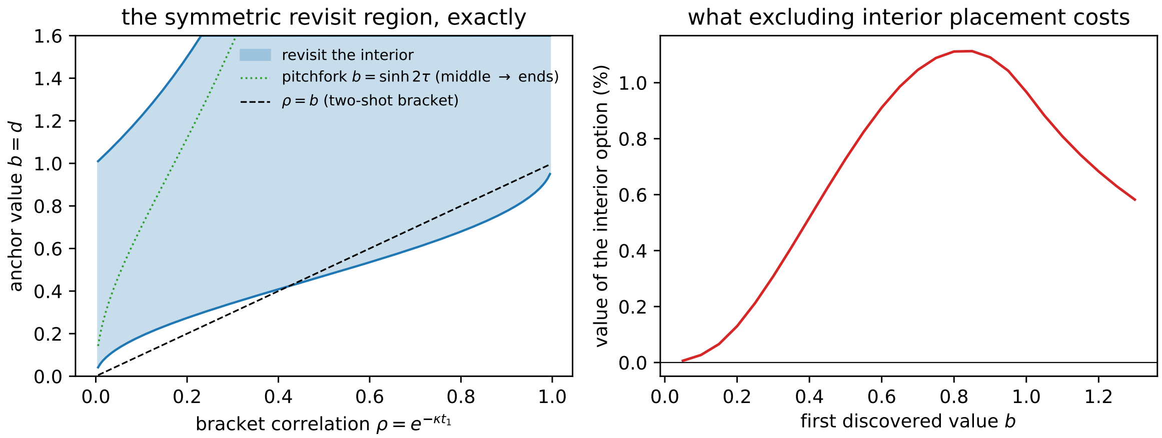

Figure 2: Left: the exact symmetric revisit region (14) in the plane of bracket correlation ρ and anchor value $b = d$, bounded below by the quadratic root b_- and above by $e^{2\Theta}$. The dotted curve is the pitchfork $b = \sinh 2\tau$: below it the optimal interior point is the bracket middle, above it the mirror pair (12) hugging the ends. The dashed line is the two-shot bracket $\rho = b$, entering the region at $b^* = 0.4233$; the upper boundary continues beyond the plotted range. Right: the value of the interior option — the percentage increase in expected payoff of the full three-shot optimum over the best policy denied interior placements (it may still place on either exterior ray) — as a function of the first discovered value b, with the bracket optimized in both cases.

9 The three-shot policy, computed

Writing $V(b, d, \rho)$ for the best of $\zeta(b)$, $\zeta(d)$ and the interior optimum (via (18)), the three-shot value of a bracket ρ is $\log \mathbb{E}_d\big[e^{V(b,d,\rho)}\big]$ with $d \sim N\big(b\rho, 1-\rho^2\big)$, computed by adaptive quadrature and optimized over ρ. The comparison policy denies interior placement but keeps both exterior rays. It is not a ban on reversing direction.

Three observations, all in Table 1 and Figure 2. First, a below-median start still means flight, and the interior option is then worthless. Second, the three-shot bracket is wider than the two-shot step, but most of the widening is *not* bought by the interior option: at $b = 0.5$ the two-shot policy steps to $\rho = 0.5$, excluding interior placement already widens the optimum to $\rho = 0.323$, and admitting it widens further to $\rho = 0.289$. The extra evaluation and the choice it enables account for most of the stretch; the interior option adds a further, smaller pull.

Third, the interior option's worth peaks near 1.1% of expected value around $b \approx 0.83$ and vanishes at both extremes. Excluding interior placement is thus cheap by value; it is the prescription that errs, and for equal anchors it errs exactly on the revisit window (14) — high, but below the ceiling $e^{2\Theta}$ above which both policies stay.

For unequal observations the erring region is a joint condition on (b, d, ρ), decided by the quartic (18), not membership of each value separately: at $\rho = \frac{1}{2}$, $b = \frac{1}{2}$, $d = 2$, both values lie in the symmetric window, yet staying at the second anchor is optimal and the exclusion costs nothing.

b	optimal ρ	value	value, interior excluded	interior worth
0.1	0.053	0.8396	0.8393	0.03%
0.3	0.167	0.8723	0.8692	0.31%
0.5	0.289	0.9386	0.9314	0.73%
0.7	0.402	1.0375	1.0271	1.05%
0.83	0.469	1.1183	1.1072	1.12%
0.9	0.502	1.1670	1.1562	1.09%
1.2	0.622	1.4059	1.3991	0.68%

Table 1: The three-shot game: optimal bracket correlation, log-value, the log-value with interior placement excluded (exterior rays in both directions permitted), and the percentage the interior option adds to expected payoff. Adaptive quadrature over the second value d; interior optima exact via the stationarity quartic (18). For $b < 0$ the optimal bracket correlation tends to zero and the limiting log-value is exactly $\frac{1}{2} + \log\bigl(1 + 1/\sqrt{2\pi}\bigr) = 0.8357\ldots$

10 The rule as a line search

Derivative-free optimizers spend most of their budget on one-dimensional subproblems: pick a direction, place a few evaluations along it, return a point. Classical inner loops (golden-section search, Brent's method) assume smooth unimodal slices. We implemented the three-evaluation rule as an inner loop — the exact final placement of Sections 5–8, the opening step from Section 9, with the mean, scale, and correlation length estimated online from the optimizer's own history — and benchmarked it inside a common outer loop (iterated random-direction search, budget 120 evaluations, matched directions and seeds) against golden-section, Brent, and random placement.

The objectives are the application suite of the HumpDay derivative-free optimization package (Cotton 2021). Each is a small physics or engineering problem in which a point in the unit cube sets the controls of a deterministic simulator and the objective is the outcome: a break in a game of pool, the pins felled in bowling, the range of a trebuchet, the layout of a wind farm. We use the 22 dynamics-simulating problems. Their one-dimensional slices range from smooth to collision-cascade rough, which is why they exercise a line search across the regimes the theory distinguishes; roughness is measured per slice by the exponent of the small-lag variogram.

The rule wins on two kinds of objective. On collision and agent-chaos objectives with rough but structured slices it is the best inner loop; on a plinko objective it beats every rival on 20 or 21 of 24 seeds. It also wins on high-dimensional objectives even when slices are smooth, a sixteen-dimensional wind-farm layout on 21 or 22 of 24 seeds.

The high-dimensional win comes from evaluation economy. The rule's one- or two-evaluation lines let the outer loop try several times more directions than an inner loop that spends eight to ten evaluations per line. The win is budget-starved rather than general: it appears when the dimension times a classical line search's cost exceeds the whole budget, so a method with a fixed number of directions per iteration cannot complete one pass.

The rule works only in this pairing with a cheap-line outer loop. It is not a drop-in replacement for the line search inside an existing optimizer. Substituted into Powell, whose direction updates rely on genuine line minimization, it loses to Brent.

The rule loses in two regimes. On landscapes too rough to carry navigable structure, broad sampling wins and placement theory has nothing to exploit. On needle landscapes, where the value hides in a spike, the fixed full-segment probes of golden-section search are the right move. On genuinely smooth slices the classical loops are preferable in principle, though among these physics objectives the smooth-slice case is rarer than reduced-order surrogates suggest, since the real collision and ballistics simulators are rough.

Across the 22 objectives the rule ranks first or second for nine. As a standalone optimizer, rather than an inner loop, the two-shot variant is mid-field against a panel of standard derivative-free methods, with mean rank near the middle of eleven. Its relative standing is best at the smallest budgets, around forty evaluations, and slips as the budget grows and population methods such as CMA-ES overtake it. That is the expected place for a rule derived from a few-evaluation terminal-placement model.

We ran a control on exact exponentiated-OU fields, the model's own landscape. The rule loses there; broad probing wins. The reported wins come from the rule's evaluation economy and its online step-size adaptation, not from the objectives resembling an OU path.

This fits the theory. A terminal-placement payoff rewards correlated trials, which the bracket supplies. A running-maximum payoff over many calls rewards spread, since a stationary field holds many nearly independent regions, so the correlated placement is partly redundant. The policy is optimal for terminal placement and we do not claim it for many-call optimization.

The rule helps when the evaluation budget is comparable to the number of local minima a slice presents. When the minima greatly outnumber the budget, spread wins. A pre-registered out-of-sample test bears this out: on 33 further HumpDay pages held out from the objectives that suggested the pattern, the rule's rank correlates with a slice's local-minimum count in the predicted direction (Spearman 0.46, one-sided $p = 0.004$; stronger with the single highest-count page, which runs against the pattern, removed). Scripts, seeds, the pre-registration, and per-objective results are in the repository.

11 The rules in business terms

The correlation length sets the regime. A searcher who can place trials many correlation lengths apart should probe far and take the best; the interior premium is first order in the bracket correlation while the value of a fresh draw is not. A venture is usually tethered. Capabilities, customers, and identity keep the next position within roughly one correlation length of the incumbent, and that is the regime in which the results above carry the decision.

The rules are short. Abandon a below-median incumbent for fresh ground. If the trial lands at or below the median, leave past the better of the two positions rather than repair the bracket (Appendix A); a trial only somewhat below the anchors can still be worth keeping. If both positions are good, reasonably balanced, and below the ceiling, commit between them: at the midpoint for moderate values, closer to the better-known position for high ones. If either position is exceptional, or the two are very uneven, stay at the best observed. The interior option is worth about one percent at the optimum, measured against a policy already free to leave in either direction, so most of the value of the last pivot is the information it produces, and the case for making the trial at all is much stronger than the case for any particular refinement afterward.

12 Discussion

Two distinct comparisons should not be merged. Against a *visited* point of matching conditional mean, any unvisited location wins by Jensen alone: it carries variance, the known value carries none. Between two *unvisited* rivals of matching mean, an interior point and its constructed exterior counterpart, Theorem 1 gives the constant variance ratio $e^{2\Theta}$. Neither comparison guarantees that a matched-mean unvisited point exists when the observed value is high.

Mean reversion caps the conditional mean of every unvisited point. Above $b = e^{2\Theta}$ the cap binds and the optimal final location is one already seen. Below it, the optimal location is past the better end when the news is bad, and between the anchors when the news is good.

A related mid-game rule is proved in Appendix A: a trial that lands at or below the process median is abandoned rather than repaired, and more generally the stay-or-leave decision after an interior trial is a threshold rule in the revealed value, with the measured threshold below the anchors.

With one evaluation remaining there is nothing to explore, since no discovery at the final point can inform a later choice. The last move is pure placement. The preference for uncertain locations comes from Jensen's inequality, not from any value of information.

Exploration proper enters only with more shots remaining. One might guess a two-phase policy, stepping out while values climb and then refining between the two best, but a direct implementation of that heuristic on rough one-dimensional landscapes showed no systematic gain over applying the two-shot rule repeatedly, so we advance no conjecture about its form. The k-shot dynamic program is well defined and each policy evaluation is cheap through the bridge formulas; the adaptive optimization it requires remains to be analyzed.

The same placement problem arises as the line-search step inside derivative-free optimizers, where one- dimensional slices of real objectives can be rough; we pursue that use separately in the humpday benchmark package.

The problem solved here is terminal placement: the payoff is the value at the final location, not the running maximum of the path. Section 2 locates it in the literature; the short version is that the tradition's exact objects are thresholds in $\sigma^2/|\mu|$ on drifting Brownian paths with endogenous evaluations and flow payoffs, and the fixed-budget, terminal-payoff, mean-reverting combination, with its phase diagram, constant amplification $e^{2\Theta}$, and constrained parabola (10), is new.

Reproducibility

All numbers and the figure are produced by `numerics2.py` alongside this source; the review-driven identities (the constant variance ratio, the $d = 1$ factorization, the stationarity quartic, the exact flee limit) are independently checked in `verify/check_review.py`, and the earlier counterexample machinery lives in `verify/check_inside.py` and `verify/check_middle.py`, all in the browniansearch repository.

A The abandonment threshold

Suppose a bracket has known end log-values b and d and an interior trial reveals log-value v. The searcher may continue inside one of the two sub-brackets or leave past an end.

Proposition 2 (A bad draw sends the searcher out). Let $v \leq 0$, the process median or below. Every placement interior to the two sub-brackets, and the revisit of v, is weakly dominated by exiting to $\zeta(\max(b,d))$; strictly if $v < 0$.

Proof. Consider the sub-bracket with near end b and far end v, and an interior point with correlations λ_2 and λ to its ends, $\rho_1 = \lambda_2\lambda$. The conditional mean is $\mu = Ab + Bv$ with $A = \lambda_2(1-\lambda^2)/(1-\rho_1^2)$ and $B = \lambda(1-\lambda_2^2)/(1-\rho_1^2)$, both nonnegative, and $A \leq \lambda_2$ because $\lambda \geq \rho_1$ implies $1-\lambda^2 \leq 1-\rho_1^2$. Since $v \leq 0$, $\mu \leq \lambda_2 b$. The conditional variance obeys $\nu = (1-\lambda_2^2)(1-\lambda^2)/(1-\rho_1^2) \leq 1-\lambda_2^2$ by the same inequality. Hence $\mu+\nu/2 \leq \lambda_2 b + (1-\lambda_2^2)/2 \leq \sup_\ell[\ell b + (1-\ell^2)/2] = \zeta(b)$, where the last identification holds for every real b.

For $v < 0$ any strictly interior point has $B > 0$, so $\mu < \lambda_2 b$. The revisit of v pays $v \leq 0 < \frac{1}{2} \leq \zeta(\max(b,d))$. The same bound applies to the other sub-bracket with d in place of b. □

Proposition 3 (Monotone value). The value of the game is nondecreasing in every observed value.

Proof. Every candidate placement's conditional mean is a nonnegative linear combination of observed values, and every conditional variance depends on locations only. The final payoff is a maximum of terms that are nondecreasing in each observed value. For earlier stages, raise one observed value and couple the reveal at any placement by shifting it with its conditional mean; induction over stages preserves monotonicity through the expectation and the maximum. □

Exits are anchored at the outer values and do not depend on the interior reveal v, while continuations inside are nondecreasing in v by the monotonicity above. The stay-or-leave decision after an interior trial is therefore a threshold rule in v. The threshold lies below the anchors: at $b = d = 0.7$ and bracket correlation $\rho = 0.5$, staying is optimal exactly when $v > 0.63$. A trial somewhat worse than both anchors is still worth staying for, because the sub-bracket keeps one good end and the variance. Verification scripts accompany the paper's repository.

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